

Notes on QFT

PROBABILITY FLOOR

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Source: <https://github.com/probability569/qft-notes-schwartz>

“note to self, do this or im the biggest retard”

feynman likely

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Preface

I was debating whether I should write this set of notes. Almost everyone who understand me well enough asks me why am I learning chemistry? Before I began this course (5.111sc), I disliked chemistry. The subjective reason is due to my poor exposure to chemistry with teachers who would say “just remember it” for exceptional cases of rules, which is rather frequent in chemistry. The objective reason is the lack of generality of chemistry rules, they seem often not fundamental enough.

However, 5.111sc does a great job on quenching my contempt for chemistry by attempting to derive most rules from first principles and going back to them when the rule doesn't work. Also, AI's have gotten good enough to answer my questions (eg. why does silver nitrate stain? can three atoms form a bond?). The professor is also convincing in her argument of the usefulness of a rule that's not the most general and fundamental: “I like simple things that works 90% of the time.”

This set of notes is based mainly on the class of 5.111sc from MIT OpenCourseWare [1]. Specifically, lectures by Prof. Catherine Drennan, recorded in 2018, and problems from its problem set. When the source of an example problem is not otherwise stated, it may be assumed to be from the problem sets from the course material.

Topics of study in this course, by virtue of MIT Registrar's Office, is as following. “Introduction to chemistry, with emphasis on basic principles of atomic and molecular electronic structure, thermodynamics, acid-base and redox equilibria, chemical kinetics, and catalysis. Introduction to the chemistry of biological, inorganic, and organic molecules.” [2]

Emphasis would be placed on explaining and predicting phenomena, especially counterintuitive ones, from mathematical and/or physical first principles. This set of notes is, by design, lighter than the mathematical notes, such as Notes on Linear Algebra [3], as most well-known results, to the extent of those being in the AP curriculum, are skimmed through.

Although officially, this course has no prerequisites, basic chemistry knowledge, math, and physics are assumed to a limited extent.

1 Atoms

Say one is stuck on a deserted island with natives having no prior scientific knowledge, but are logical, what series of experiments and theories does one must state to create a chain of reasons to convince the natives of every chemistry facts one knows?

This chapter wishes to accomplish such task for chemistry involving a single atom, though arguably not chemical.

1.1 Existence of Atoms and Atomic Structure

From Dalton's Law of Definite Proportions and the Greek's cutting argument, it becomes rather convincing that there is a small unit of matter. Given a microscope, Brownian motion further verifies the existence of such smallest unit of matter, define it as an *atom*.

Via radioactive material or Hydrogen gas in plasma phase, subatomic particles can be found to have charge and their charge-to-mass ratio may be determined through mass spectrometry. The Oil Drop Experiment [4] gives the charge. Hence, the mass is known. Rutherford's experiment of bombarding gold foil with α particles from RaBr_2 then shows the existence of positively charged nuclei and its size relative to the atom. The size of an atom may be found through X-ray Crystallography, by diffraction to get the distance between two nuclei. The charge-to-mass ratio of an electron (or β particle) then reveals the mass of the electron.

However, Newtonian mechanics would give the speed of the electron to be faster than light, which is impossible from Maxwell's equations. The break-through point lies in the ultraviolet catastrophe motivates the postulate of quantized energy, which gives rise to planck's constant. The same constant is then observed in the photoelectric effect as

$$E = h\nu, \quad (1.1.1)$$

and is verified by the fact that the excitation of the electrons dependent on the frequency but not intensity of the incident light. This assumption gives the particle-like description of light, the *photon*; however, Thomas Young showed in the Double Slit Experiment that light is also a particle. The particle light duality of light and the equation for its momentum $p = \frac{E}{\lambda}$ motivates the wave description for matter.

1.2 The Schrödinger Equation

Further developments in quantum mechanics, most notably the Schrödinger equation. The time-dependent Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi \quad (1.2.1)$$

describes how a quantum state changes with time, where i is the imaginary unit ($i^2 = -1$), $\hbar$ is the reduced Planck constant, Ψ is the system's wave function, t is time, $\frac{\partial \Psi}{\partial t}$ is the wave function's rate of change with time, and $\hat{H}$ is the Hamiltonian operator representing the system's total energy. For electrons, the wave function $\Psi(r, \theta, \varphi)$ is parametrized by its position relative to the nucleus. One may observe the resemblance of the equation to the defining equation of eigenvalues and eigenvectors of matrices. In fact, the eigenvalues of $\hat{H}$ correspond to the allowed electron energies and the eigenfunctions corresponding wave functions or orbitals, the magnitude of the square of the wave function Ψ is the probabilistic distribution of the electron. For a single nonrelativistic particle, the Hamiltonian is $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + U(r)$, where $-\frac{\hbar^2}{2m}\nabla^2$ is the kinetic-energy operator, m is the particle's mass, ∇^2 is the Laplacian describing spatial curvature, and $U(r)$ is the particle's potential energy at position r .

The generality of the Schrödinger equation and its accuracy with observations (on the order of magnitude of $10^{-10}\%$) makes it extraordinarily useful in describing the electron cloud of atoms. Hence, to an exaggerated extent, *chemistry is about approximating the Schrödinger equation in complicated scenarios.*

The Schrödinger eigenvalue equation for hydrogen is

$$\hat{H}\Psi = E\Psi.$$

For the energy eigenstate labelled by n ,

$$\hat{H}\Psi = -\frac{1}{n^2} \frac{me^4}{8\epsilon_0^2 \hbar^2} \Psi.$$

Define the Rydberg energy by

$$R_H = \frac{me^4}{8\epsilon_0^2 \hbar^2} \approx 2.180 \times 10^{-18} \text{ J} \approx 13.606 \text{ eV}.$$

Therefore,

$$E_n = -\frac{1}{n^2} \frac{me^4}{8\epsilon_0^2 \hbar^2} = -\frac{R_H}{n^2}, \quad n = 1, 2, 3, \dots$$

If binding energy means the positive energy required to ionize the atom, then

$$B_n = -E_n = \frac{R_H}{n^2}. \quad (1.2.2)$$

Generalizing, the binding energy equation for any one-electron atom is

$$E = 13.6 \frac{Z^2}{n^2} eV,$$

where Z is the atomic number and n is the principal quantum number.

From (1.2.2) and (1.1.1), the wavelength of the photon omitted when an electron moves from the principal quantum number n_i to n_f is

$$\nu = \frac{R_H}{h} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

1.3 Atomic Orbitals

Eigenfunctions from (1.2.1) gives wave function for a given nucleus to be described by three quantum numbers.

The principal quantum number n is a positive integer that determines the binding energy of an electron assuming single electron (1.2.2). In this case of multiple electrons, there may be a correction term due to electron repulsion that is relatively small compared to the term given by (1.2.2). Hence, categorize all electrons in an atom by this number gives the shell of the electron.

The orbital angular momentum quantum number ℓ takes value from $\ell = 0, 1, 2, \dots, n-1$. It determines the angular-momentum magnitude and the subshell or orbital shape.

The magnetic quantum number m (or m_ℓ) takes values from $m = -\ell, -\ell + 1, \dots, \ell - 1, \ell$. It determines $L_z = m\hbar$ and labels a particular orbital within a subshell.

The wave function is then given by

$$\Psi_{n\ell m}(r, \theta, \varphi).$$

An orbital is given the three defining quantum number mentioned above, denote it with a number concatenated by $\overline{n\ell m_\ell}$.

Bibliography

- [1] C. Drennan, “Principles of Chemical Science.” [Online]. Available: <https://ocw.mit.edu/courses/5-111sc-principles-of-chemical-science-fall-2014/>
- [2] Massachusetts Institute of Technology, “Fall 2026 Course 5: Chemistry.” Accessed: Aug. 29, 2026. [Online]. Available: <https://student.mit.edu/catalog/m5a.html>
- [3] Saint Even, Slipper King, and Icy Cheees, “Notes on Linear Algebra.” [Online]. Available: <https://yunfeix2009.github.io/lin-alg-notes-ocw/>
- [4] Wikipedia contributors, “Oil Drop Experiment.” Aug. 2026.